TASK SPECIFICATION: P7-01 PHASE P7 — NOTIFICATIONS, GOVERNANCE & TELEMETRY ANALYTICS

In-App Notification Engine & Notification API

Git Branch: `feature/P7-01-notifications-api` · Target Pull Request: feature/P7-01-notifications-api

Core Objective & Technical Scope:
Real-time notification service managing unread alerts, batch marking as read, and persistent event notifications.

Priority Level	High	Complexity Rating	`M`
Execution Status	Backlog 📋	Target PR / Commit	`feature/P7-01-notifications-api`
Upstream Dependencies	P2-03		
Architecture Blockers	None (Ready for execution)		

IMPLEMENTATION ROADMAP & ENGINEERING STEPS

1. Author V11__notifications.sql migration creating in_app_notification table.
2. Implement Notification domain entity, repository, and service.
3. Implement NotificationController with mark-as-read and unread count endpoints.
4. Enforce security checks ensuring users access only their own notifications.

TARGET FILE INVENTORY & EXPECTED ARTIFACTS

- `backend/src/main/resources/db/migration/V11__notifications.sql`
- `notification/**`

ACCEPTANCE CRITERIA & QUALITY CHECKLIST

- Notification queries return only recipient records with zero data leakage.
- Unread count query optimized for frequent polling.

VERIFICATION & TEST COVERAGE STRATEGY

- NotificationServiceTest
- NotificationControllerTest
- SecurityIsolationTest